

KAYLASTUY, Russian Federation

WMO: 309780

Lat: 49.833N

Lon: 118.383E

Elev: 550

StdP: 94.89

Time Zone: 9.00 (E09)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.5	-35.0	-40.2	0.1	-37.2	-37.6	0.1	-34.8	10.6	-15.7	9.3	-16.6	0.7	70	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	31.4	18.8	29.5	18.0	27.7	17.6	20.6	27.4	19.8	25.9	19.0	24.7	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.4	22.7	17.8	13.6	22.3	16.8	12.8	22.0	62.4	27.5	59.5	26.1	56.6	24.7	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.3	8.1	-38.8	34.8	4.1	3.2	-41.8	37.1	-44.2	39.0	-46.6	40.8	-49.6	43.1		
			-39.2	22.2	4.2	1.2	-42.2	23.1	-44.7	23.8	-47.0	24.4	-50.1	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-25.1	-20.1	-8.7	3.3	11.8	18.6	21.2	19.0	11.8	1.7	-11.3	-22.0
	DBStd	17.00	5.32	5.56	6.70	5.01	4.38	3.83	2.80	3.32	4.28	4.80	6.62	5.61
	HDD10.0	4666	1088	842	579	209	30	1	0	0	29	257	639	991
	HDD18.3	6848	1347	1075	838	452	207	41	6	32	198	514	889	1249
	CDD10.0	1065	0	0	0	7	86	260	347	279	84	2	0	0
	CDD18.3	206	0	0	0	0	5	50	95	52	4	0	0	0
	CDH23.3	1928	0	0	0	3	99	564	777	429	56	1	0	0
	CDH26.7	608	0	0	0	0	27	196	251	123	12	0	0	0
Wind	WSAvg	3.2	2.0	2.5	3.5	4.5	4.4	3.4	3.0	2.9	3.2	3.3	2.9	2.2
Precipitation	PrecAvg	291	3	3	5	9	21	53	85	64	30	13	5	6
	PrecMax	516	14	9	19	31	77	120	225	177	61	52	16	26
	PrecMin	172	0	0	0	0	1	6	6	16	1	0	0	0
	PrecStd	84	3	2	5	8	18	30	52	43	17	13	5	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.0	-2.5	10.9	22.5	29.9	35.0	33.7	32.6	28.2	18.0	6.9	-5.5
		MCWB	-9.9	-4.9	3.7	11.3	15.4	18.1	19.9	20.0	16.3	9.2	2.2	-7.1
	2%	DB	-12.6	-6.0	6.8	18.1	26.2	31.0	31.6	29.9	24.6	14.5	3.2	-9.7
		MCWB	-13.4	-7.7	1.2	8.6	13.9	17.5	19.5	19.0	14.6	7.1	-0.5	-10.9
	5%	DB	-15.3	-8.5	4.0	15.1	23.3	28.8	29.3	27.8	22.0	11.9	0.3	-12.0
		MCWB	-16.0	-9.8	-0.4	6.7	12.5	17.1	18.7	18.1	13.3	5.8	-2.4	-12.9
	10%	DB	-17.3	-11.0	1.4	12.1	20.8	26.6	27.5	25.7	19.6	9.3	-2.3	-14.0
		MCWB	-18.0	-12.0	-1.9	5.0	11.2	16.3	18.3	17.4	12.2	4.3	-4.3	-14.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.1	-5.0	4.0	11.3	16.5	20.5	22.1	21.4	17.2	9.7	2.3	-7.1
		MCDB	-9.0	-2.8	10.2	21.5	27.5	29.9	29.4	29.4	24.7	16.7	6.6	-5.6
	2%	WB	-13.4	-7.7	1.6	9.0	14.6	19.2	21.0	20.2	15.8	7.7	-0.3	-10.8
		MCDB	-12.6	-6.0	6.4	17.8	24.7	26.8	27.7	26.7	22.0	13.3	2.7	-9.9
	5%	WB	-15.9	-9.8	-0.2	7.1	13.2	18.1	20.2	19.3	14.4	6.2	-2.3	-12.8
		MCDB	-15.3	-8.5	3.7	14.5	22.1	25.7	26.3	25.0	20.5	11.3	0.1	-12.0
	10%	WB	-17.9	-12.0	-1.9	5.3	11.7	17.2	19.5	18.5	12.9	4.6	-4.2	-14.7
		MCDB	-17.3	-11.0	1.4	11.8	19.7	24.6	25.2	23.7	18.8	9.0	-2.3	-14.1
Mean Daily Temperature Range	5% DB	MDBR	10.4	12.5	12.6	12.5	12.8	11.9	10.5	11.1	12.3	11.3	10.7	9.9
		MCDBR	11.4	13.5	14.2	17.6	17.3	15.8	14.1	14.7	16.1	14.9	12.5	11.2
		MCWBR	10.7	11.9	9.8	10.3	8.6	6.3	5.3	6.2	8.1	9.1	9.1	10.2
	5% WB	MCDBR	11.2	13.5	13.4	16.7	15.8	12.6	10.9	11.4	13.5	13.7	12.0	11.1
		MCWBR	10.6	12.0	9.6	10.2	8.3	5.9	5.0	5.4	7.0	8.8	9.0	10.1
Clear-Sky Solar Irradiance	taub	0.258	0.281	0.331	0.419	0.494	0.476	0.428	0.398	0.348	0.321	0.287	0.266	
	taud	2.219	2.175	2.118	2.013	1.864	2.034	2.248	2.299	2.371	2.340	2.264	2.179	
	Ebn at Noon	787	856	865	818	770	790	826	836	846	801	738	707	
	Edh at Noon	72	98	126	158	192	163	130	118	98	84	70	66	
All-Sky Solar Radiation	RadAvg	1.68	2.92	4.47	5.39	6.08	6.33	5.86	5.32	4.19	2.84	1.70	1.23	
	RadStd	0.12	0.16	0.22	0.32	0.39	0.49	0.40	0.41	0.26	0.15	0.09	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)		N/A	-2.05	-2.75	N/A	N/A	N/A	+211	+200	N/A	N/A

Nomenclature: See separate page